

A Thesis Proposal Title Submitted to the Department of Artificial Intelligence, School of Electrical Engineering and Computing

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Compute-Efficient Morphology-Aware Retrieval and Entity Resolution for Ge’ez Script Question Answering Systems

Research Description

Standard Retrieval-Augmented Generation (RAG) pipelines lose retrieval recall across modern Ethio-Semitic languages written in the Ge’ez script (Amharic and Tigrinya)—often dropping below 40% Recall@20 on BM25—while imposing heavy computational demands during indexing and inference. This performance loss is driven by non-concatenative Semitic root-and-pattern inflections, attached clitics (fused prepositions such as *ke-/kab-*, *be-/bi-*, *le-/ni-* and conjunctions *ina-/mis-*), and orthographic homophones (laryngeal *hä/ḥä/ḫä*, sibilant *sä/śä*, glottal *ʾä/ʿä*, and ejective *ṣä/ṣ́ä*), which cause identical semantic entities to be indexed as disparate tokens or fractured graph nodes. Conversely, commercial graph-based RAG frameworks (such as LightRAG) rely on costly proprietary LLM API calls, rendering them economically and computationally impractical for low-resource environments and local workstation deployments in Ethiopia.

This research designs a compute-efficient, morphology-aware clitic segmentation and entity-resolution module for Ge’ez-script question answering that operates strictly on local workstation hardware. We develop a unified, lightweight morphological segmenter that strips peripheral clitics and normalizes homophones across both Amharic and Tigrinya without destroying grammatical stems, dramatically reducing dictionary explosion. To ensure alignment with dense representation geometries, we evaluate morphological normalization both zero-shot and with retriever fine-tuning across three paradigms: sparse lexical representations (splade-geez-base), dense bi-encoders (bge-m3), and token-level late-interaction search (colbert-geez-base). Furthermore, we integrate a localized dynamic entity resolution module using compact open foundation models (Qwen2.5-3B and Llama-3.2-3B) to resolve surface entity variants into unified cross-lingual knowledge graph nodes without requiring external API dependencies or high-end multi-GPU infrastructure.

The complete pipeline will be evaluated on the **AmQA** benchmark (Amharic), **TiQA** and **HornMT** (Tigrinya), and an extracted statutory corpus combining the **Ethiopian Federal Negarit Gazette** (Amharic-English) with regional legal documents under a strict single-GPU budget (24GB VRAM). Evaluation will measure passage retrieval Recall@K, Mean Reciprocal Rank (MRR), entity resolution precision, answer faithfulness, indexing throughput, and end-to-end inference latency. The results will determine the trade-off between morphological preprocessing overhead and retrieval accuracy, establishing a reproducible, low-compute RAG framework tailored for Ethiopian public and academic institutions.